Chapter 10: Derivatives of Multivariable Functions (§10.1 - 10.4)

Instructor: Shengwen Guo

Department of Mathematics, St. Bonaventure Univ.

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Contents

Preview Activity

We start with the function f of one variable:

limits of functions of one variable

- $f(x) = 3 - x$, what is $\lim_{x \to 0} f(x)$?
- $f(x) = |x|$, what is $\lim_{x \to 0} f(x)$?
- $f(x) = \frac{|x|}{x}$, what is $\lim_{x \to 0} f(x)$?

Example 10.1.1

Let $f(x, y) = 3 - x - 2y$. Plot the graph and level curves of $f(x, y)$, what do they suggest about

$$\lim_{(x,y)\to(0,0)} f(x, y)?$$

Limits of single variable calculus

Intuitively, we say that a function f has a limit L as x approaches a provided that we can make the values $f(x)$ as close to L as we like by taking x sufficiently close (but not equal) to a. We denote this behavior by writing

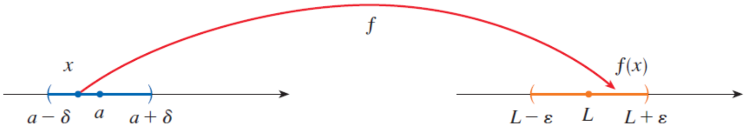
$$\lim_{x \to a} f(x) = L.$$

If we translate it into math language: For any $\varepsilon > 0$, there exists a $\delta > 0$, such that

$$\text{if } \; |x - a| < \delta, \; \text{then } \; |f(x) - f(a)| < \varepsilon.$$

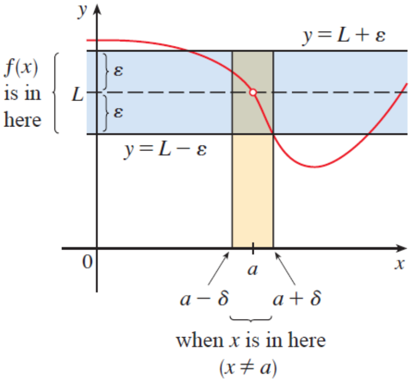
Interpretation

The definition of limit says that if any small interval $(L-\varepsilon, L+\varepsilon)$ is given around L, then we can find an interval $(a-\delta, a+\delta)$ around a such that f maps all the points in $(a-\delta, a+\delta)$ (except possibly a) into the interval $(L-\varepsilon, L+\varepsilon)$.



Another geometric interpretation

If $\varepsilon > 0$ is given, then we draw the horizontal lines $y = L - \varepsilon$ and $y = L + \varepsilon$ and the graph of f. If $\lim_{x \to a} f(x) = L$, then we can (always) find a number $\delta > 0$ (for every ε) s.t. if we restrict x to lie in the interval $(a - \delta, a + \delta)$ and take $x \neq a$, the the curve $y = f(x)$ lies between the lines $y = L - \varepsilon$ and $y = L + \varepsilon$.



Example 10.1.2

Let $f(x,y) = \dfrac{\sin(x^2+y^2)}{x^2+y^2}$. What does the table below suggest about $\lim\limits_{(x,y)\to(0,0)} f(x,y)$?

$x \backslash y$	−1.0	−0.5	−0.2	0	0.2	0.5	1.0
−1.0	0.455	0.759	0.829	0.841	0.829	0.759	0.455
−0.5	0.759	0.959	0.986	0.990	0.986	0.959	0.759
−0.2	0.829	0.986	0.999	1.000	0.999	0.986	0.829
0	0.841	0.990	1.000		1.000	0.990	0.841
0.2	0.829	0.986	0.999	1.000	0.999	0.986	0.829
0.5	0.759	0.959	0.986	0.990	0.986	0.959	0.759
1.0	0.455	0.759	0.829	0.841	0.829	0.759	0.455

Example 10.1.2

Let $f(x,y) = \dfrac{x^2 - y^2}{x^2 + y^2}$. What does the table below suggest about $\displaystyle\lim_{(x,y)\to(0,0)} f(x,y)$?

$x \backslash y$	-1.0	-0.5	-0.2	0	0.2	0.5	1.0
-1.0	0.000	0.600	0.923	1.000	0.923	0.600	0.000
-0.5	-0.600	0.000	0.724	1.000	0.724	0.000	-0.600
-0.2	-0.923	-0.724	0.000	1.000	0.000	-0.724	-0.923
0	-1.000	-1.000	-1.000		-1.000	-1.000	-1.000
0.2	-0.923	-0.724	0.000	1.000	0.000	-0.724	-0.923
0.5	-0.600	0.000	0.724	1.000	0.724	0.000	-0.600
1.0	0.000	0.600	0.923	1.000	0.923	0.600	0.000

Definition: Limits of functions of two variables

Given a function $f = f(x, y)$, we say that f has limit L as (x, y) approaches (a, b) provided that we can make $f(x, y)$ as close to L as we like by taking (x, y) sufficiently close (but not equal) to (a, b). We write

$$\lim_{(x,y)\to(a,b)} f(x, y) = L$$

Another detailed definition

Let $D_\delta((a,b))$ be the disk with center (a,b) and radius $\delta > 0$. Let $f = f(x,y)$ whose domain D includes points arbitrarily close to (a,b). Then we say that

$$\lim_{(x,y)\to(a,b)} f(x,y) = L$$

if for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$\text{if } (x,y) \in D \text{ and } (x,y) \in D_\delta((a,b)) \text{ then } |f(x,y) - L| < \varepsilon$$

The existence of limits

- To investigate the (existence) of limit of a single variable function, $\lim_{x \to a} f(x)$, we often consider the behavior of f as x approaches a from right and from the left.
- Similarly, we may investigate limits of two-variable functions, $\lim_{(x,y)\to(a,b)} f(x,y)$ by considering the behavior of f as (x,y) approaches (a,b) from various directions.

Activity 10.1.2. Consider the function f, defined by

$$f(x, y) = \frac{y}{\sqrt{x^2 + y^2}},$$

whose graph is shown below in Figure 10.1.18

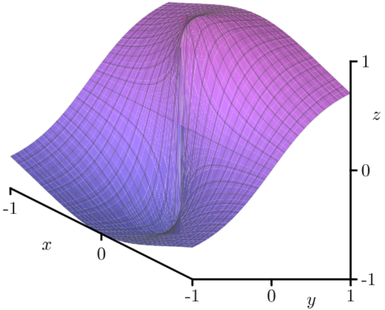


Figure 10.1.18. The graph of $f(x, y) = \frac{y}{\sqrt{x^2+y^2}}$.

a. Is f defined at the point $(0,0)$? What, if anything, does this say about whether f has a limit at the point $(0,0)$?

b. Values of f (to three decimal places) at several points close to $(0,0)$ are shown in Table 10.1.19.

Table 10.1.19. Values of a function f.

$x\backslash y$	-1.000	-0.100	0.000	0.100	1.000
-1.000	-0.707	—	0.000	—	0.707
-0.100	—	-0.707	0.000	0.707	—
0.000	-1.000	-1.000	—	1.000	1.000
0.100	—	-0.707	0.000	0.707	—
1.000	-0.707	—	0.000	—	0.707

Based on these calculations, state whether f has a limit at $(0,0)$ and give an argument supporting your statement. (Hint: The blank spaces in the table are there to help you see the patterns.)

c. Now we formalize the conjecture from the previous part by considering what happens if we restrict our attention to different paths. First, we look at f for points in the domain along the x-axis; that is, we consider what happens when $y = 0$. What is the behavior of $f(x, 0)$ as $x \to 0$? If we approach $(0, 0)$ by moving along the x-axis, what value do we find as the limit?

d. What is the behavior of f along the line $y = x$ when $x > 0$; that is, what is the value of $f(x, x)$ when $x > 0$? If we approach $(0, 0)$ by moving along the line $y = x$ in the first quadrant (thus considering $f(x, x)$ as $x \to 0^+$), what value do we find as the limit?

e. In general, if $\lim_{(x,y)\to(0,0)} f(x, y) = L$, then $f(x, y)$ approaches L as (x, y) approaches $(0, 0)$, regardless of the path we take in letting $(x, y) \to (0, 0)$. Explain what the last two parts of this activity imply about the existence of $\lim_{(x,y)\to(0,0)} f(x, y)$.

f. Shown below in Figure 10.1.20 is a set of contour lines of the function f. What is the behavior of $f(x,y)$ as (x,y) approaches $(0,0)$ along any straight line? How does this observation reinforce your conclusion about the existence of $\lim_{(x,y)\to(0,0)} f(x,y)$ from the previous part of this activity? (Hint: Use the fact that a non-vertical line has equation $y = mx$ for some constant m.)

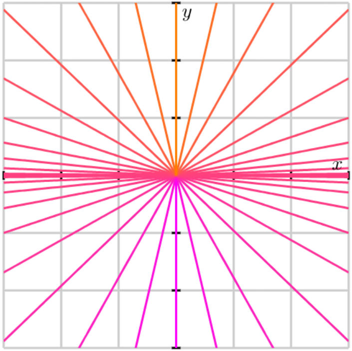


Figure 10.1.20. Contour lines of $f(x,y) = \frac{y}{\sqrt{x^2+y^2}}$.

From the previous example and an example in Calculus I we know that

- In the one variable example $g(x) = \dfrac{x}{|x|}$, since

$$\lim_{x \to 0^-} g(x) \neq \lim_{x \to 0^+} g(x),$$

  $\lim\limits_{x \to 0} g(x)$ does not exist.

- When $f(x,y) = \dfrac{y}{\sqrt{x^2+y^2}}$, since

$$\lim_{x \to 0, y=0} f(x,y) \neq \lim_{x=0, y \to 0} f(x,y),$$

  we conclude that $\lim\limits_{(x,y) \to (0,0)} f(x,y)$ does not exist.

As a general rule, we have

Limits along different paths

If $f(x, y)$ has two different limits as (x, y) approaches (a, b) along two different paths, then $\lim_{(x,y)\to(a,b)} f(x, y)$ does not exist.

Note: We can NEVER conclude that the limit of a function exists only by considering its behavior along different paths.

Activity 10.1.3. Let's consider the function g defined by

$$g(x,y) = \frac{x^2y}{x^4+y^2}$$

and investigate the limit $\lim_{(x,y)\to(0,0)} g(x,y)$.

a. What is the behavior of g on the x-axis? That is, what is $g(x,0)$ and what is the limit of g as (x,y) approaches $(0,0)$ along the x-axis?

b. What is the behavior of g on the y-axis? That is, what is $g(0,y)$ and what is the limit of g as (x,y) approaches $(0,0)$ along the y-axis?

c. What is the behavior of g on the line $y = mx$? That is, what is $g(x, mx)$ and what is the limit of g as (x,y) approaches $(0,0)$ along the line $y = mx$?

d. Based on what you have seen so far, do you think $\lim_{(x,y)\to(0,0)} g(x,y)$ exists? If so, what do you think its value is?

e. Now consider the behavior of g on the parabola $y = x^2$? What is $g(x,x^2)$ and what is the limit of g as (x,y) approaches $(0,0)$ along this parabola?

f. State whether the limit $\lim_{(x,y)\to(0,0)} g(x,y)$ exists or not and provide a justification of your statement.

Example

Consider $f(x, y) = \dfrac{x^2y^2}{x^2 + y^2}$. Does the limit $\lim\limits_{(x,y)\to(0,0)} f(x, y)$ exist?

- If limit exists, what is the possible value?
- As $0 \leq x^2 \leq x^2 + y^2$, so how can we estimate and simplify $f(x, y)$?

Properties of Limits.

Let $f = f(x, y)$ and $g = g(x, y)$ be functions so that $\lim_{(x,y)\to(a,b)} f(x, y)$ and $\lim_{(x,y)\to(a,b)} g(x, y)$ both exist. Then

1. $\displaystyle\lim_{(x,y)\to(a,b)} x = a$ and $\displaystyle\lim_{(x,y)\to(a,b)} y = b$
2. $\displaystyle\lim_{(x,y)\to(a,b)} cf(x, y) = c\left(\lim_{(x,y)\to(a,b)} f(x, y)\right)$ for any scalar c
3. $\displaystyle\lim_{(x,y)\to(a,b)} [f(x, y) \pm g(x, y)] = \lim_{(x,y)\to(a,b)} f(x, y) \pm \lim_{(x,y)\to(a,b)} g(x, y)$
4. $\displaystyle\lim_{(x,y)\to(a,b)} [f(x, y)g(x, y)] = \left(\lim_{(x,y)\to(a,b)} f(x, y)\right)\left(\lim_{(x,y)\to(a,b)} g(x, y)\right)$
5. $\displaystyle\lim_{(x,y)\to(a,b)} \frac{f(x, y)}{g(x, y)} = \frac{\displaystyle\lim_{(x,y)\to(a,b)} f(x, y)}{\displaystyle\lim_{(x,y)\to(a,b)} g(x, y)}$ if $\displaystyle\lim_{(x,y)\to(a,b)} g(x, y) \neq 0.$

Example

Does $\lim_{(x,y)\to(0,0)}$ exists for the following given $f(x,y)$?

- $f(x,y) = \dfrac{xy}{x^2+y^2}$
- $f(x,y) = \dfrac{xy^2}{x^2+y^4}$
- $f(x,y) = \dfrac{3x^2y}{x^2+y^2}$
- $f(x,y) = 3x^2y^3 + 2xy^2 - 3x + 1$

Continuity of univariable functions

A function f of a single variable f is said to be continuous at $x = a$ provided that the following three conditions are satisfied:

1. $f(a)$ exists
2. $\lim_{x \to a} f(x)$ exists, and
3. $\lim_{x \to a} f(x) = f(a)$

Continuity

Definition: continuity of bivariate functions

A function $f(x, y)$ is continuous at (a, b) provided that

1. $f(a, b)$ exists
2. $\displaystyle\lim_{(x,y)\to(a,b)} f(x, y)$ exists, and
3. $\displaystyle\lim_{(x,y)\to(a,b)} f(x, y) = f(a, b)$

Note:

- All polynomial functions of two variables are continuous on $\mathbb{R}^2$.
- Any rational function (ratio of polynomials) is continuous on its domain.

Example

Evaluate

$$\lim_{(x,y)\to(1,2)} (x^2y^3 - x^3y^2 + 3x + 2y)$$

Example

Let

$$g(x,y) = \begin{cases} \frac{x^2-y^2}{x^2+y^2} & \text{if } (x,y) \neq (0,0) \\ 0 & \text{if } (x,y) = (0,0) \end{cases}.$$

Is $g(x,y)$ continuous everywhere on $\mathbb{R}^2$?

Functions of three or more variables

Everything that we have done in this section can be extended to functions of three or more variables.

Definition: limits of $f = f(x, y, z)$

Let $B_\delta((a, b, c))$ be the ball with center (a, b, c) and radius $\delta > 0$. Let $f = f(x, y, z)$ whose domain D includes points arbitrarily close to (a, b, c). Then we say that

$$\lim_{(x,y,z)\to(a,b,c)} f(x, y, z) = L$$

if for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$\text{if } (x, y, z) \in D \text{ and } (x, y, z) \in B_\delta((a, b, c)) \text{ then } |f(x, y, z) - L| < \varepsilon$$

And similarly, The function f is continuous at (a, b, c) if

$$\lim_{(x,y,z)\to(a,b,c)} f(x, y, z) = f(a, b, c)$$

If we use vector notation, we can write the definitions of a limit for functions of two or three variables in a single compact form as follows.

Limits of multivariable functions

If f is defined on a subset D of $\mathbb{R}^n$, then $\lim_{\mathbf{x} \to \mathbf{a}} f(\mathbf{x}) = L$ means that for every $\varepsilon > 0$ there is a corresponding number $\delta > 0$ such that

$$\text{if } \mathbf{x} \in D \text{ and } 0 < |\mathbf{x} - \mathbf{a}| < \delta \text{ then } |f(\mathbf{x}) - L| < \varepsilon$$

In this case the definition of continuity can be written as

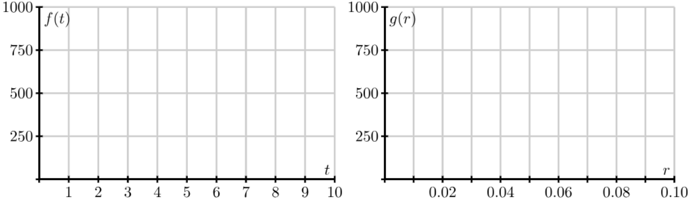
$$\lim_{\mathbf{x} \to \mathbf{a}} f(\mathbf{x}) = f(\mathbf{a})$$

Preview Activity 10.2.1. Suppose we take out an \$18,000 car loan at interest rate r and we agree to pay off the loan in t years. The monthly payment, in dollars, is

$$M(r,t) = \frac{1500r}{1 - \left(1 + \frac{r}{12}\right)^{-12t}}.$$

a. What is the monthly payment if the interest rate is 3% so that $r = 0.03$, and we pay the loan off in $t = 4$ years?

b. Suppose the interest rate is fixed at 3%. Express M as a function f of t alone using $r = 0.03$. That is, let $f(t) = M(0.03, t)$. Sketch the graph of f on the left of Figure 10.2.1. Explain the meaning of the function f.



Preview Activity

c. Find the instantaneous rate of change $f'(4)$ and state the units on this quantity. What information does $f'(4)$ tell us about our car loan? What information does $f'(4)$ tell us about the graph you sketched in (b)?

d. Express M as a function of r alone, using a fixed time of $t = 4$. That is, let $g(r) = M(r, 4)$. Sketch the graph of g on the right of Figure 10.2.1. Explain the meaning of the function g.

e. Find the instantaneous rate of change $g'(0.03)$ and state the units on this quantity. What information does $g'(0.03)$ tell us about our car loan? What information does $g'(0.03)$ tell us about the graph you sketched in (d)?

First-Order partial derivatives

Example

Consider the function $f(x, y) = \dfrac{x^2 \sin(2y)}{32}$, which measures the range, or horizontal distance, in feet, traveled by a projectile launched with an initial speed of x feet per second at an angle y radians to the horizontal.

- Fix the angle $y = 0.6$, plot the trace $f(x, 0.6)$.
- Find $\dfrac{d}{dx} f(x, 0.6)$, when $x = 150$, find $\dfrac{d}{dx} f(x, 0.6)|_{x=150}$.
- Fix the speed $x = 150$, plot the trace $f(150, y)$.
- Find $\dfrac{d}{dy} f(150, y)$, when $y = 0.6$, find $\dfrac{d}{dy} f(150, y)|_{y=0.6}$.

By holding y fixed and differentiating with respect to x, we obtain the first-order partial derivative of f with respect to x. Denoting this partial derivative as f_x, we have seen that

$$f_x(150, 0.6) = \frac{d}{dx} f(x, 0.6)|_{x=150} = \lim_{h \to 0} \frac{f(150 + h, 0.6) - f(150, 0.6)}{h}.$$

Similarly, we fix x and differentiate w.r.t y, obtaining the first-order partial derivative of f with respect to y. Denoting this partial derivative as f_y, we have seen that

$$f_y(150, 0.6) = \frac{d}{dy} f(150, y)|_{y=0.6} = \lim_{h \to 0} \frac{f(150, 0.6 + h) - f(150, 0.6)}{h}.$$

Definition: First order derivatives

The first-order partial derivatives of f with respect to x and y at a point (a, b) are, respectively,

$$f_x(a,b) = \lim_{h \to 0} \frac{f(a+h,b) - f(a,b)}{h}$$
$$f_y(a,b) = \lim_{h \to 0} \frac{f(a,b+h) - f(a,b)}{h},$$

provided the limits exist.

Activity 10.2.2. Consider the function f defined by

$$f(x, y) = \frac{xy^2}{x+1}$$

at the point $(1, 2)$.

a. Write the trace $f(x, 2)$ at the fixed value $y = 2$. On the left side of Figure 10.2.5, draw the graph of the trace with $y = 2$ around the point where $x = 1$, indicating the scale and labels on the axes. Also, sketch the tangent line at the point $x = 1$.

b. Find the partial derivative $f_x(1,2)$ and relate its value to the sketch you just made.

c. Write the trace $f(1,y)$ at the fixed value $x=1$. On the right side of Figure 10.2.5, draw the graph of the trace with $x=1$ indicating the scale and labels on the axes. Also, sketch the tangent line at the point $y=2$.

d. Find the partial derivative $f_y(1,2)$ and relate its value to the sketch you just made.

As these examples show, each partial derivative at a point arises as the derivative of a one-variable function defined by fixing one of the coordinates. Due to the connection between one-variable derivatives and partial derivatives, we have

Notations for partial derivatives

$$f_x(x,y) = f_x = \frac{\partial f}{\partial x} = \frac{\partial}{\partial x} f(x,y) = \frac{\partial z}{\partial x} = f_1 = D_1 f = D_x f$$
$$f_y(x,y) = f_y = \frac{\partial f}{\partial y} = \frac{\partial}{\partial y} f(x,y) = \frac{\partial z}{\partial y} = f_2 = D_2 f = D_y f$$

Rule for Finding Partial Derivatives of $z = f(x, y)$

1. To find f_x, regard y as a constant and differentiate $f(x, y)$ with respect to x.
2. To find f_y, regard x as a constant and differentiate $f(x, y)$ with respect to y.

Partial derivatives can also be defined for functions of three or more variables. For example, if f is a function of three variables x, y, and z, then its partial derivative with respect to x is defined as

$$f_x(x, y, z) = \lim_{h \to 0} \frac{f(x+h, y, z) - f(x, y, z)}{h}$$

and it is found by regarding y and z as constants and differentiating $f(x, y, z)$ with respect to x.

Partial derivatives for functions of n variables

If $u = f(x_1, x_2, \ldots, x_n)$, its partial derivative with respect to x_i is

$$\frac{\partial u}{\partial x_i} = \lim_{h \to 0} \frac{f(x_1, \ldots, x_{i-1}, x_i + h, x_{i+1}, \ldots, x_n) - f(x_1, \ldots, x_i, \ldots, x_n)}{h}$$

and we also write

$$\frac{\partial u}{\partial x_i} = \frac{\partial f}{\partial x_i} = f_{x_i} = f_i = D_i f$$

Activity 10.2.3.

a. If $f(x,y) = 3x^3 - 2x^2y^5$, find the partial derivatives f_x and f_y.

b. If $f(x,y) = \dfrac{xy^2}{x+1}$, find the partial derivatives f_x and f_y.

c. If $g(r,s) = rs\cos(r)$, find the partial derivatives g_r and g_s.

d. Assuming $f(w,x,y) = (6w+1)\cos(3x^2 + 4xy^3 + y)$, find the partial derivatives f_w, f_x, and f_y.

e. Find all possible first-order partial derivatives of $q(x,t,z) = \dfrac{x2^t z^3}{1+x^2}$.

Recall that the derivative of a single variable function has a geometric interpretation as the slope of the line tangent to the graph at a given point. Similarly, we have seen that the partial derivatives measure the slope of a line tangent to a trace of a function of two variables as shown in Figure 10.2.6.

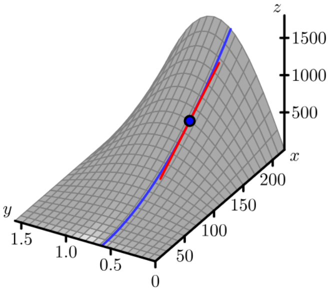


Figure 10.2.6. Tangent lines to two traces of the distance function.

Activity 10.2.4. The speed of sound C traveling through ocean water is a function of temperature, salinity and depth. It may be modeled by the function

$$C = 1449.2 + 4.6T - 0.055T^2 + 0.00029T^3 + (1.34 - 0.01T)(S - 35) + 0.016D.$$

Here C is the speed of sound in meters/second, T is the temperature in degrees Celsius, S is the salinity in grams/liter of water, and D is the depth below the ocean surface in meters.

a. State the units in which each of the partial derivatives, C_T, C_S and C_D, are expressed and explain the physical meaning of each.
b. Find the partial derivatives C_T, C_S and C_D.
c. Evaluate each of the three partial derivatives at the point where $T = 10$, $S = 35$ and $D = 100$. What does the sign of each partial derivatives tell us about the behavior of the function C at the point $(10, 35, 100)$?

Recall that for a single-variable function f, the second derivative of f is defined to be the derivative of the first derivative. That is,

$$f''(x) = \frac{d}{dx}[f'(x)] = \lim_{h \to 0} \frac{f'(x+h) - f'(x)}{h}$$

Preview Activity 10.3.1. Once again, let's consider the function f defined by $f(x,y) = \frac{x^2 \sin(2y)}{32}$ that measures a projectile's range as a function of its initial speed x and launch angle y. The graph of this function, including traces with $x = 150$ and $y = 0.6$, is shown in Figure 10.3.1 .

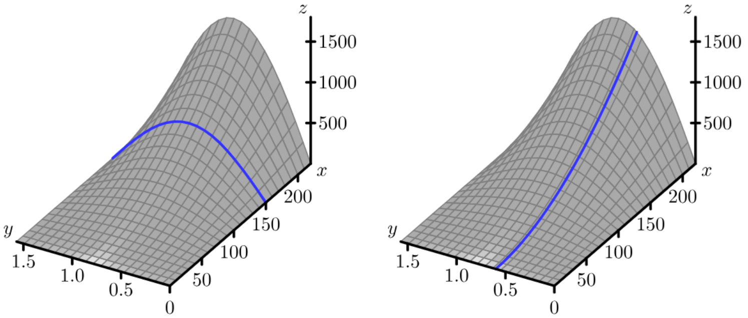


Figure 10.3.1. The distance function with traces $x = 150$ and $y = 0.6$.

a. Compute the partial derivative f_x. Notice that f_x itself is a new function of x and y, so we may now compute the partial derivatives of f_x. Find the partial derivative $f_{xx} = (f_x)_x$ and show that $f_{xx}(150, 0.6) \approx 0.058$.

b. Figure 10.3.2 shows the trace of f with $y = 0.6$ with three tangent lines included. Explain how your result from part (a) of this preview activity is reflected in this figure.

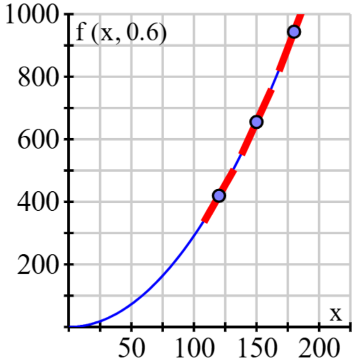


Figure 10.3.2. The trace with $y = 0.6$.

c. Determine the partial derivative f_y, and then find the partial derivative $f_{yy} = (f_y)_y$. Evaluate $f_{yy}(150, 0.6)$.

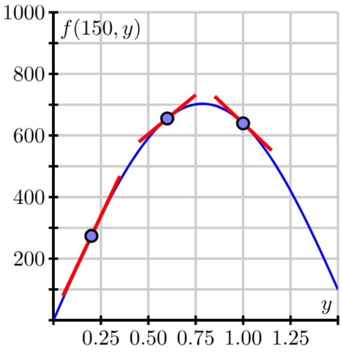


Figure 10.3.3. More traces of the range function.

d. Figure 10.3.3 shows the trace $f(150, y)$ and includes three tangent lines. Explain how the value of $f_{yy}(150, 0.6)$ is reflected in this figure.

e. Because f_x and f_y are each functions of both x and y, they each have two partial derivatives. Not only can we compute $f_{xx} = (f_x)_x$, but also $f_{xy} = (f_x)_y$; likewise, in addition to $f_{yy} = (f_y)_y$, but also $f_{yx} = (f_y)_x$. For the range function $f(x, y) = \frac{x^2 \sin(2y)}{32}$, use your earlier computations of f_x and f_y to now determine f_{xy} and f_{yx}. Write one sentence to explain how you calculated these "mixed" partial derivatives.

Second-Order Partial Derivatives

Notice that

- A function f of two independent variables x and y has two first order partial derivatives, f_x and f_y;
- Both f_x and f_y has two partial derivatives, giving a total of four second-order partial derivatives:

Second-Order Partial Derivatives

- $f_{xx} = (f_x)_x = \dfrac{\partial}{\partial x}\left(\dfrac{\partial f}{\partial x}\right) = \dfrac{\partial^2 f}{\partial x^2}$;
- $f_{yy} = (f_y)_y = \dfrac{\partial}{\partial y}\left(\dfrac{\partial f}{\partial y}\right) = \dfrac{\partial^2 f}{\partial y^2}$;
- $f_{xy} = (f_x)_y = \dfrac{\partial}{\partial y}\left(\dfrac{\partial f}{\partial x}\right) = \dfrac{\partial^2 f}{\partial y \partial x}$;
- $f_{yx} = (f_y)_x = \dfrac{\partial}{\partial x}\left(\dfrac{\partial f}{\partial y}\right) = \dfrac{\partial^2 f}{\partial x \partial y}$;

The first two are called unmixed second-order partial derivatives while the last two are called the mixed second-order partial derivatives.

Activity 10.3.2. Find all second order partial derivatives of the following functions. For each partial derivative you calculate, state explicitly which variable is being held constant.

a. $f(x, y) = x^2y^3$
b. $f(x, y) = y\cos(x)$
c. $g(s, t) = st^3 + s^4$
d. How many second order partial derivatives does the function h defined by $h(x, y, z) = 9x^9z - xyz^9 + 9$ have? Find h_{xz} and h_{zx} (you do not need to find the other second order partial derivatives).

Clairaut's Theorem

Notice that $f_{xy} = f_{yx}$ in activity 10.3.2. This is not just a coincidence. It turns out that the mixed partial derivatives f_{xy} and f_{yx} are equal for most functions that one meets in practice. The following theorem, which was discovered by the French mathematician Alexis Clairaut (1713–1765), gives conditions under which we can assert that $f_{xy} = f_{yx}$.

Clairaut's Theorem

Suppose f is defined on a disk D that contains the point (a, b). If the functions f_{xy} and f_{yx} are both continuous on D, then

$$f_{xy}(a, b) = f_{yx}(a, b)$$

Higher order partial derivatives

Partial derivatives of order 3 or higher can also be defined. For instance,

$$f_{xyy} = (f_{xy})_y = \frac{\partial}{\partial y}\left(\frac{\partial^2 f}{\partial y \partial x}\right) = \frac{\partial^3 f}{\partial y^2 \partial x}$$

Example

Calculate f_{xxyz} if

$$f(x, y, z) = \sin(3x + yz)$$

Interpreting the Second-Order Partial Derivatives

Example

Consider $f(x, y) = \sin(x)e^{-y}$. Figure 10.3.4 shows the graph of this function along with the trace given by $y = -1.5$. Also shown are three tangent lines to this trace, with increasing x-values from left to right among the three plots.

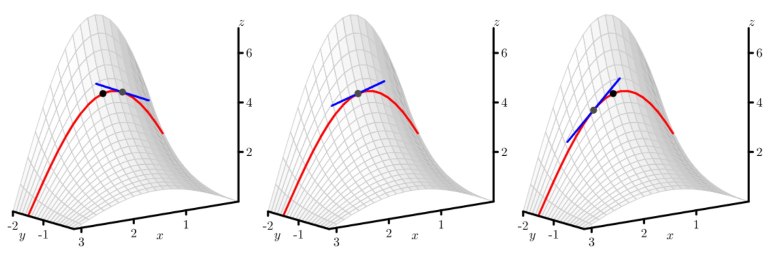


Figure 10.3.4. The tangent lines to a trace with increasing x.

- We find that the slope of tangent line (to the trace $f(x, 1.5)$) is decreasing as x increases;
- It reflects that the trace is concave down;
- Indeed, we see that

$$f_x(x, y) = \cos(x)e^{-y}, \quad f_{xx}(x, y) = -\sin(x)e^{-y} < 0$$

Activity 10.3.3. We continue to consider the function f defined by $f(x, y) = \sin(x)e^{-y}$.

a. In Figure 10.3.5, we see the trace of $f(x, y) = \sin(x)e^{-y}$ that has x held constant with $x = 1.75$. We also see three different lines that are tangent to the trace of f in the y direction at values of y that are increasing from left to right in the figure. Write a couple of sentences that describe whether the slope of the tangent lines to this curve increase or decrease as y increases, and, after computing $f_{yy}(x, y)$, explain how this observation is related to the value of $f_{yy}(1.75, y)$. Be sure to address the notion of concavity in your response.(You need to be careful about the directions in which x and y are increasing.)

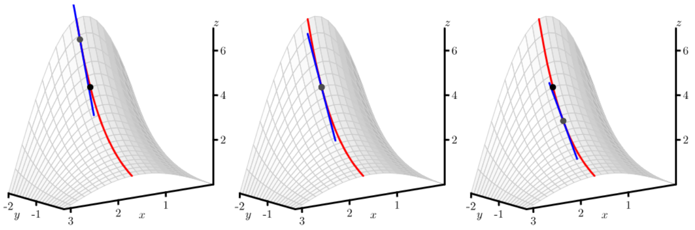


Figure 10.3.5. The tangent lines to a trace with increasing y.

b. In Figure 10.3.6, we start to think about the mixed partial derivative, f_{xy}. Here, we first hold y constant to generate the first-order partial derivative f_x, and then we hold x constant to compute f_{xy}. This leads to first thinking about a trace with x being constant, followed by slopes of tangent lines in the x-direction that slide along the original trace. You might think of sliding your pencil down the trace with x constant in a way that its slope indicates $(f_x)_y$ in order to further animate the three snapshots shown in the figure.

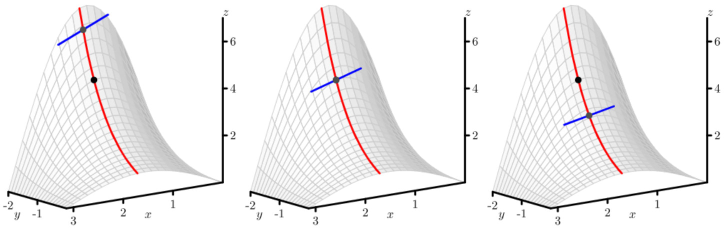


Figure 10.3.6. The trace of $z = f(x, y) = \sin(x)e^{-y}$ with $x = 1.75$, along with tangent lines in the y-direction at three different points.

Based on Figure 10.3.6, is $f_{xy}(1.75, -1.5)$ positive or negative? Why?

c. Determine the formula for $f_{xy}(x, y)$, and hence evaluate $f_{xy}(1.75, -1.5)$. How does this value compare with your observations in (b)?

d. We know that $f_{xx}(1.75, -1.5)$ measures the concavity of the $y = -1.5$ trace, and that $f_{yy}(1.75, -1.5)$ measures the concavity of the $x = 1.75$ trace. What do you think the quantity $f_{xy}(1.75, -1.5)$ measures?

e. On Figure 10.3.6, sketch the trace with $y = -1.5$, and sketch three tangent lines whose slopes correspond to the value of $f_{yx}(x, -1.5)$ for three different values of x, the middle of which is $x = -1.5$. Is $f_{yx}(1.75, -1.5)$ positive or negative? Why? What does $f_{yx}(1.75, -1.5)$ measure?

In summary,

- f_{xx} is the second derivative of a trace of f where $y = y_0$ is fixed, it measures the concavity of $f(x, y_0)$. Similarly, f_{yy} is the second derivative of a trace of f where $x = x_0$ is fixed, it measures the concavity of $f(x_0, y)$.
- f_{xy} measures the "twist" of the graph as we increase y along a particular trace where x is held constant. In the same way, f_{yx} measures how the graph twists as we increase x.

- Recall that in single variable calculus, the graph of a differentiable function, when viewed on a very small scale, looks like a line. We call this line the **tangent line**, and its slope is given by the derivative.
- We can develop this idea in $\mathbb{R}^3$. As we view the graph of a two-variable function on a small scale, it looks like a plane, named the **tangent plane**.

Example

Let's consider $f(x, y) = 6 - \frac{x^2}{2} - y^2$. We choose to study the behavior of this function near the point $(x_0, y_0) = (1, 1)$. In particular, we wish to view the graph on an increasingly small scale around this point, as shown in the two plots in Figure 10.4.2.

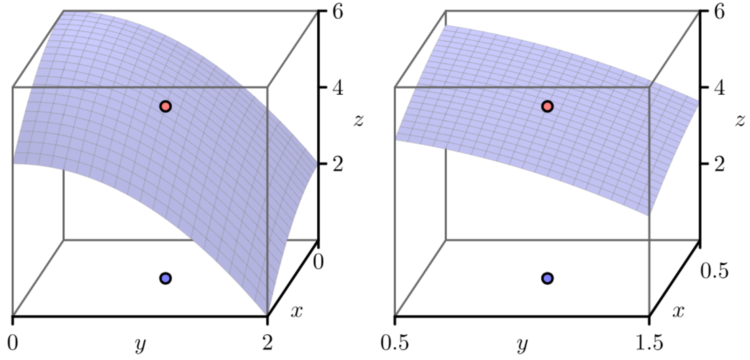


Figure 10.4.2. The graph of $f(x, y) = 6 - x^2/2 - y^2$.

As figure 10.4.3 shows, when viewed on a small scale, we see that the graph of this particular two-variable function looks like a plane.

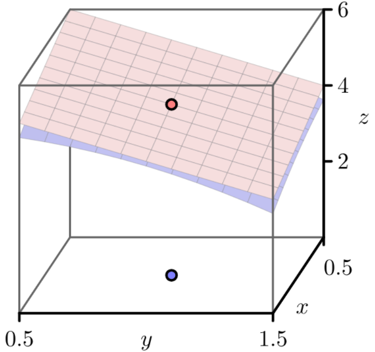


Figure 10.4.3. The graph of $f(x, y) = 6 - x^2/2 - y^2$.

Preview Activity 10.4.1. Let $f(x, y) = 6 - \frac{x^2}{2} - y^2$, and let $(x_0, y_0) = (1, 1)$.

a. Evaluate $f(x, y) = 6 - \frac{x^2}{2} - y^2$ and its partial derivatives at (x_0, y_0); that is, find $f(1, 1)$, $f_x(1, 1)$, and $f_y(1, 1)$.

b. We know one point on the tangent plane; namely, the z-value of the tangent plane agrees with the z-value on the graph of $f(x, y) = 6 - \frac{x^2}{2} - y^2$ at the point (x_0, y_0). In other words, both the tangent plane and the graph of the function f contain the point (x_0, y_0, z_0). Use this observation to determine z_0 in the expression $z = z_0 + a(x - x_0) + b(y - y_0)$.

c. Sketch the traces of $f(x,y) = 6 - \frac{x^2}{2} - y^2$ for $y = y_0 = 1$ and $x = x_0 = 1$ below in Figure 10.4.4.

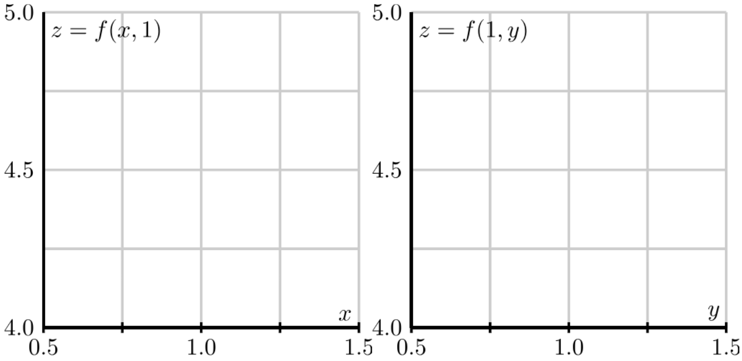


Figure 10.4.4. The traces of $f(x,y)$ with $y = y_0 = 1$ and $x = x_0 = 1$.

d. Determine the equation of the tangent line of the trace that you sketched in the previous part with $y = 1$ (in the x direction) at the point $x_0 = 1$.

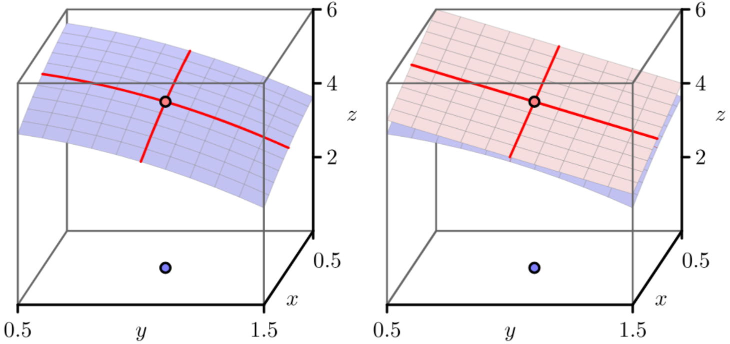


Figure 10.4.5. The traces of $f(x, y)$ and the tangent plane.

e. Figure 10.4.5 shows the traces of the function and the traces of the tangent plane. Explain how the tangent line of the trace of f, whose equation you found in the last part of this activity, is related to the tangent plane. How does this observation help you determine the constant a in the equation for the tangent plane $z = z_0 + a(x - x_0) + b(y - y_0)$? (Hint: How do you think $f_x(x_0, y_0)$ should be related to $z_x(x_0, y_0)$?)

f. In a similar way to what you did in (d), determine the equation of the tangent line of the trace with $x = 1$ at the point $y_0 = 1$. Explain how this tangent line is related to the tangent plane, and use this observation to determine the constant b in the equation for the tangent plane $z = z_0 + a(x - x_0) + b(y - y_0)$. (Hint: How do you think $f_y(x_0, y_0)$ should be related to $z_y(x_0, y_0)$?)

g. Finally, write the equation $z = z_0 + a(x - x_0) + b(y - y_0)$ of the tangent plane to the graph of $f(x, y) = 6 - x^2/2 - y^2$ at the point $(x_0, y_0) = (1, 1)$.

The Tangent Plane

The ideas of tangent plane are as follows:

- When we look at the graph of a single-variable function on a small scale near a point x_0, we expect to see a line; in this case, we say that f is **locally linear** near x_0; In single-variable calculus, we learn that if the derivative of a function exists at a point, then the function is guaranteed to be locally linear there.
- In a similar way, we say that a two-variable function $f(x, y)$ is **locally linear near** (x_0, y_0) provided that the graph of f looks like a plane (its tangent plane) when viewed on a small scale near (x_0, y_0).

Differentiability

Question

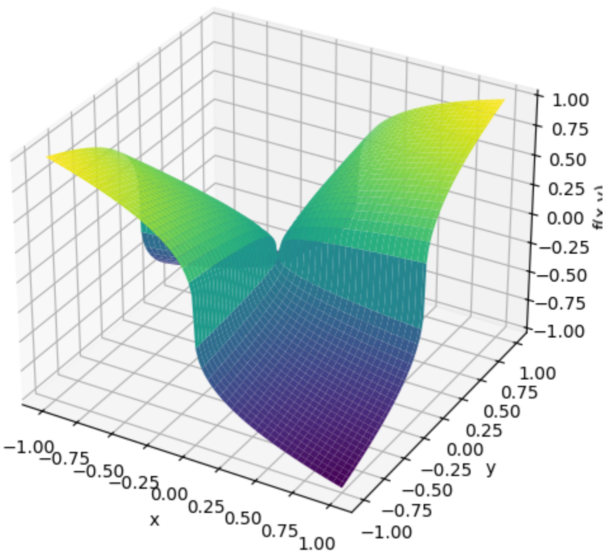
Does the existence of f_x and f_y at (x_0, y_0) guarantee that $f(x, y)$ is locally linear at (x_0, y_0)?

Example

The surface for $f(x, y) = x^{1/3}y^{1/3}$. $f_x(0, 0)$ and $f_y(0, 0)$ exist, however, $f(x, y)$ is not locally linear at $(0, 0)$.

So the existence of the two first order partial derivatives at a point does not guarantee local linearity at that point.

Graph of f(x,y) = x^{1/3}*y^{1/3}



Suppose that

- $z = f(x, y)$ is a two- variable function
- x changes from x_0 to $x_0 + \Delta x$;
- y changes from y_0 to $y_0 + \Delta y$

Then the increment of z is

$$\Delta z = f(x + \Delta x, y + \Delta y) - f(x, y).$$

And we define the differentiability of $f(x, y)$ as follows:

Differentiability

If $z = f(x, y)$, then f is differentiable at (x_0, y_0) if Δz can be expressed in the form

$$\Delta z = f_x(x_0, y_0)\Delta x + f_y(x_0, y_0)\Delta y + \varepsilon_1 \Delta x + \varepsilon_2 \Delta y$$

where $\varepsilon_1, \varepsilon_2 \to 0$ as $(\Delta x, \Delta y) \to (0, 0)$.

Now we can define a stronger, but more easily verified, conditions that ensure local linearity.

Continuously differentiability

If f is a function of the independent variables x and y and both f_x and f_y exist and are continuous in an open disk containing the point (x_0, y_0), then f is continuously differentiable at (x_0, y_0).

Whenever a function $z = f(x, y)$ is continuously differentiable at a point (x_0, y_0), it follows that the function has a tangent plane at (x_0, y_0).

Note: Basically, if f is **differentiable** in an open disk containing (x_0, y_0), then f is locally constant at (x_0, y_0). Continuous differentiability is stronger than differentiability but is easier to verify.

In Preview Activity 10.4.1, we find the following general formula for the tangent plane.

The tangent plane

If $f(x, y)$ has continuous first-order partial derivatives, then the equation of the plane tangent to the graph of f at the point $(x_0, y_0, f(x_0, y_0))$ is

$$z = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$$

Important Note

- It is possible that $f_x(x_0, y_0)$ and $f_y(x_0, y_0)$ exist for f (for example, $f(x, y) = (xy)^{1/3}$ at $(0, 0)$), and so the plane $z = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$ exists even though f is not locally linear at (x_0, y_0). In such a case this plane is not tangent to the graph.
- Differentiability for a function of two variables implies the existence of a tangent plane, but the existence of the two first order partial derivatives of a function at a point does not imply differentiaility. This is quite different than what happens in single variable calculus.

Activity 10.4.2.

a. Find the equation of the tangent plane to $f(x, y) = 2 + 4x - 3y$ at the point $(1, 2)$. Simplify as much as possible. Does the result surprise you? Explain.

b. Find the equation of the tangent plane to $f(x, y) = x^2y$ at the point $(1, 2)$.

In single variable calculus, an important use of the tangent line is to approximate the value of a differentiable function.

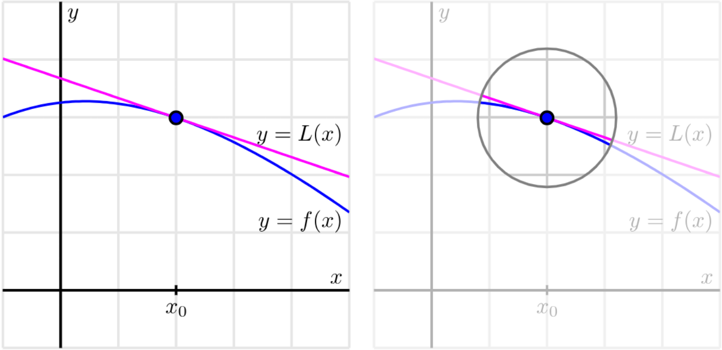


Figure 10.4.6. The linearization of the single-variable function $f(x)$.

Denote $L(x)$ the tangent line at $(x_0, f(x_0))$, its equation is

$$L(x) = f(x_0) + f'(x_0)(x - x_0).$$

Note that $L(x) \approx f(x)$ near $x = x_0$.

In the same way, the tangent plane to the graph of a differentiable function $f(x, y)$ at a point (x_0, y_0) provides a good approximation of $f(x, y)$ near (x_0, y_0). That is to say, the tangent plane at (x_0, y_0) is close to the graph of $f(x, y)$ near (x_0, y_0). We let $L(x, y)$ be the equation of tangent plane at (x_0, y_0), then

$$L(x, y) = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0).$$

And $L(x, y) \approx f(x, y)$ for points near (x_0, y_0).

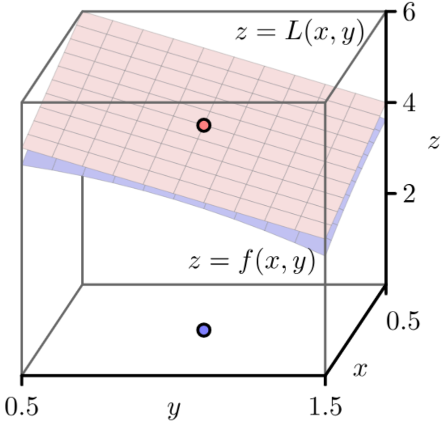


Figure 10.4.7. The linearization of $f(x, y)$.

Example

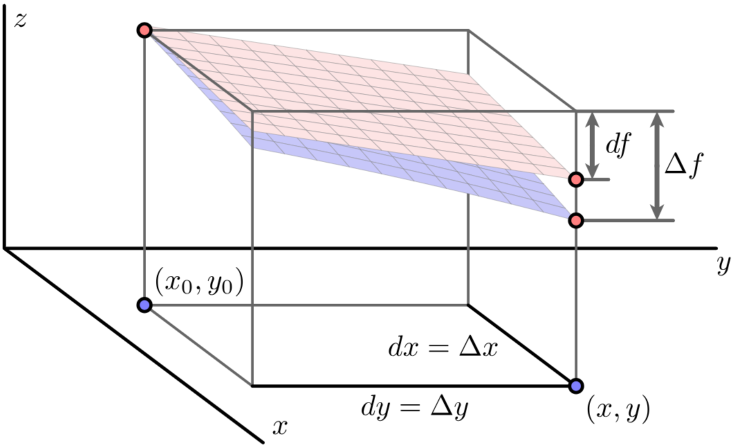
Find the linearization $L(x, y)$ for the function g defined by

$$g(x, y) = \frac{x}{x^2 + y^2}$$

at the point $(1, 2)$. Then use the linearization to estimate the value of $g(0.8, 2.3)$.

Differentials

Sometimes we are more interested in the change in f as we move from (x_0, y_0) to $(x, y) = (x_0 + \Delta x, y_0 + \Delta y)$.



Define $\Delta f = f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0)$. Based on the linearization result, we have

$$\begin{aligned}\Delta f \approx df &= L(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0) \\ &= [f(x_0, y_0) + f_x(x_0, y_0)\Delta x + f_y(x_0, y_0)\Delta y] - f(x_0, y_0) \\ &= f_x(x_0, y_0)\Delta x + f_y(x_0, y_0)\Delta y.\end{aligned}$$

Let $dx = \Delta x$ and $dy = \Delta y$, we have

$$\boxed{df = f_x(x_0, y_0)dx + f_y(x_0, y_0)dy}.$$

We call the quantities dx, dy, and df differentials, and we think of them as measuring small changes in the quantities x, y and f.

Example

Suppose we have a machine that manufactures rectangles of width $x = 20$ cm and height $y = 10$ cm. However, the machine isn't perfect, and therefore the width could be off by $dx = \Delta x = 0.2$ cm and the height could be off by $dy = \Delta y = 0.4$ cm. Let $A = A(x, y)$ is the area of the rectangle, what is dA?

Activity 10.4.4. The questions in this activity explore the differential in several different contexts.

a. Suppose that the elevation of a landscape is given by the function h, where we additionally know that $h(3,1) = 4.35$, $h_x(3,1) = 0.27$, and $h_y(3,1) = -0.19$. Assume that x and y are measured in miles in the easterly and northerly directions, respectively, from some base point $(0,0)$. Your GPS device says that you are currently at the point $(3,1)$. However, you know that the coordinates are only accurate to within 0.2 units; that is, $dx = \Delta x = 0.2$ and $dy = \Delta y = 0.2$. Estimate the uncertainty in your elevation using differentials.

b. The pressure, volume, and temperature of an ideal gas are related by the equation

$$P = P(T,V) = 8.31T/V,$$

where P is measured in kilopascals, V in liters, and T in kelvin. Find the pressure when the volume is 12 liters and the temperature is 310 K. Use differentials to estimate the change in the pressure when the volume increases to 12.3 liters and the temperature decreases to 305 K.